

Tech & Teach

Digital Chip Design Program

RTL Design

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Task #03

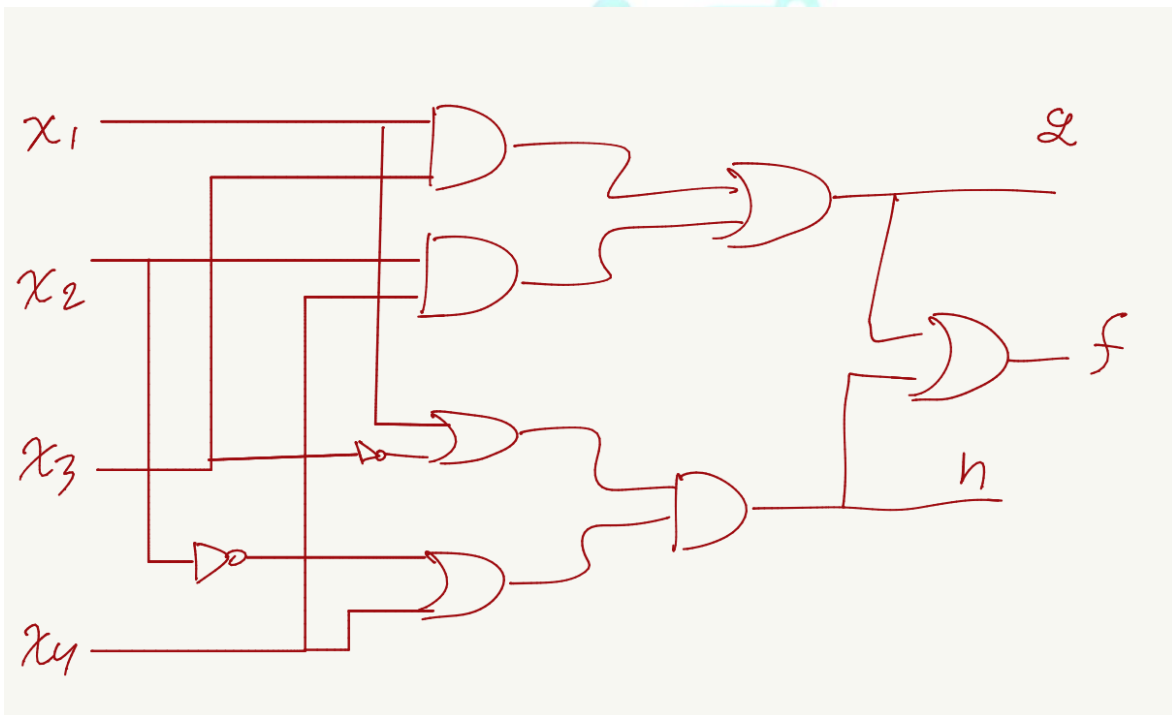
Behavioral

Tech & Teach

Task #1.1

```
module t3(f, x1, x2, x3);  
  input x1, x2, x3;  
  output reg f;  
  always_comb begin  
    if ( (x1 && x2) || (~x2 && x3) )  
      f = 1;  
    else f = 0;  
  end  
endmodule
```

Task #1.2



Task #1.3

```
module t3(AeqB, AltB, AgtB, a, b);  
input [3:0] a, b;  
output reg AeqB, AltB, AgtB;
```

```
always @(*) begin  
    if (a == b) begin  
        AeqB = 1;  
        AltB = 0;  
        AgtB = 0;  
    end  
    else if (a < b) begin  
        AeqB = 0;  
        AltB = 1;  
        AgtB = 0;  
    end  
    else if (a > b) begin  
        AeqB = 0;  
        AltB = 0;  
        AgtB = 1;  
    end  
    else begin  
        AeqB = 1'bx;  
        AltB = 1'bx;  
        AgtB = 1'bx;  
    end  
end  
end  
endmodule
```

Work distribution

Section	Abdulmalek	Omar	Khaled
Report	✓		
Task n.x		✓	
Task n.x+1			✓